



This image shows a single sheet of white paper with horizontal ruling lines. The lines are evenly spaced and run across the width of the page. There are no margins, text, or other markings on the paper.

SPRING AI

GENERATIVE ARTIFICIAL INTELLIGENCE CON JAVA

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Corso avanzato per Venis S.p.A, Venezia, Italia

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Note

This image shows a single sheet of white paper with horizontal blue or grey ruling lines. The lines are evenly spaced and run across the width of the page, providing a template for handwriting practice. There are no margins, text, or other markings on the paper.



FUNCTION AS TOOL

Note

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-  Simone Scannapieco

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```
package it.venis.ai.spring.demo.data;

public enum Unit {

    C,
    F;

}
```

```
package it.venis.ai.spring.demo.model;

import it.venis.ai.spring.demo.data.Unit;

public record WeatherRequest(String location, Unit unit) {}
```

```
package it.venis.ai.spring.demo.model;

import it.venis.ai.spring.demo.data.Unit;

public record TemperatureResponse(double temp, Unit unit) {}
```

[illegible]

```
package it.venis.ai.spring.demo.services;

import java.util.function.Function;

import it.venis.ai.spring.demo.data.Unit;
import it.venis.ai.spring.demo.model.WeatherRequest;
import it.venis.ai.spring.demo.model.TemperatureResponse;

public class TemperatureService implements Function<WeatherRequest, TemperatureResponse> {

    public TemperatureResponse apply(WeatherRequest request) {
        return new TemperatureResponse(30.0, Unit.C);
    }

}
```

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```
package it.venis.ai.spring.demo.config;

...

@Configuration
public class WeatherToolsConfig {

    public static final String GET_TEMP_IN_LOCATION_FUNCTION_NAME = "getTemperatureInLocation";

    ToolCallback toolCallback = FunctionToolCallback
        .builder(GET_TEMP_IN_LOCATION_FUNCTION_NAME, new TemperatureService())
        .description("Ottieni la temperatura corrente nella località specificata.")
        .inputType(WeatherRequest.class)
        .toolMetadata(ToolMetadata.builder()
            .returnDirect(true)
            .build())
        .build();

    @Bean
    public ChatClient geminiWeatherToolsChatClient(OpenAiChatModel geminiChatModel) {

        ChatClient.Builder chatClientBuilder = ChatClient.builder(geminiChatModel);
        return chatClientBuilder
            .defaultToolCallbacks(toolCallback)
            .defaultAdvisors(List.of(new SimpleLoggerAdvisor()))
            .defaultSystem(
                """
                Sei un assistente AI di nome WeatherGeminiBot, addestrato per fornire
                informazioni meteo alle persone.
                Utilizza quando possibile il tool 'getTemperatureInLocation'.
                """)
            .build();
    }

    ...
}
```

Note

[illegible]

```
...
@Bean
public ChatClient ollamaWeatherToolsChatClient(OllamaChatModel ollamaChatModel) {

    ChatClient.Builder chatClientBuilder = ChatClient.builder(ollamaChatModel);

    return chatClientBuilder
        .defaultToolCallbacks(toolCallback)
        .defaultAdvisors(List.of(new SimpleLoggerAdvisor(), new OllamaCostSavingsAdvisor()))
        .defaultSystem(
            """
            Sei un assistente AI di nome WeatherLlamaBot, addestrato per fornire
            informazioni meteo alle persone.
            """)
        .build();
}
}
```

[illegible]


```
package it.venis.ai.spring.demo.controllers;

...

@RestController
@Configuration
public class QuestionController {

    private final QuestionService service;
    private final RAGService ragService;
    private final TimeToolsService timeToolsService;

    private final ChatClient geminiWeatherToolsChatClient;
    private final ChatClient ollamaWeatherToolsChatClient;

    public QuestionController(QuestionService service,
                             RAGService ragService,
                             TimeToolsService timeToolsService,
                             @Qualifier("geminiWeatherToolsChatClient") ChatClient geminiWeatherToolsChatClient,
                             @Qualifier("ollamaWeatherToolsChatClient") ChatClient ollamaWeatherToolsChatClient) {

        this.service = service;
        this.ragService = ragService;
        this.timeToolsService = timeToolsService;
        this.geminiWeatherToolsChatClient = geminiWeatherToolsChatClient;
        this.ollamaWeatherToolsChatClient = ollamaWeatherToolsChatClient;
    }

    ...
}
```

[illegible]

```

...
@PostMapping("/gemini/ask/weather-tools/temperature")
public TemperatureResponse getGeminiTemperatureToolAnswer(@RequestBody WeatherRequest request) {

    return this.geminiWeatherToolsChatClient
        .prompt()
        .call()
        .entity(TemperatureResponse.class);
}

@PostMapping("/ollama/ask/weather-tools/temperature")
public TemperatureResponse getOllamaTemperatureToolAnswer(@RequestBody WeatherRequest request) {

    return this.ollamaWeatherToolsChatClient
        .prompt()
        .call()
        .entity(TemperatureResponse.class);
}
}

```

[illegible]

Branch: 13-spring-ai-gemini-ollama-weather-tools

[illegible]